

LEBANON, NH, USA

WMO: 726116

Lat: 43.626N

Lon: 72.305W

Elev: 182

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94765

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-19.3	-27.4	0.3	-21.1	-24.5	0.4	-17.7	8.9	-1.9	8.1	-3.8	0.9	350	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	31.8	21.8	30.1	21.0	28.4	20.1	23.5	29.0	22.6	27.6	21.7	26.2	3.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.0	26.2	21.1	16.1	25.2	20.1	15.1	24.4	70.9	29.0	67.3	27.5	64.0	26.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.1	5.9	DB	-27.9	34.5	3.4	1.5	-30.4	35.6	-32.4	36.4	-34.3	37.2	-36.7	38.3	(4)
			WB	-28.1	25.3	3.3	0.9	-30.5	26.0	-32.5	26.5	-34.3	27.0	-36.8	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-6.8	-5.3	-0.2	7.0	13.7	18.3	21.2	20.2	16.3	9.3	3.1	-3.0
	DBStd	10.84	6.68	5.95	5.49	4.79	4.30	3.69	2.95	3.03	3.97	4.51	4.56	5.67
	HDD10.0	2082	519	429	317	116	13	0	0	0	2	71	211	404
	HDD18.3	4067	778	662	574	342	156	46	8	16	87	283	456	662
	CDD10.0	1312	0	0	2	25	127	249	348	318	190	47	5	1
	CDD18.3	255	0	0	0	1	11	44	98	75	24	1	0	0
	CDH23.3	2541	0	0	3	39	196	481	947	654	211	11	0	0
	CDH26.7	786	0	0	1	11	56	149	326	186	58	0	0	0
Wind	WSAvg	2.0	2.1	2.2	2.5	2.5	2.2	1.9	1.8	1.6	1.6	1.9	2.0	1.9
Precipitation	PrecAvg	905	62	53	62	69	86	75	81	85	82	79	87	77
	PrecMax	1241	151	167	106	130	227	157	182	254	161	177	173	181
	PrecMin	596	11	2	10	28	28	26	15	34	14	7	35	25
	PrecStd	168	40	28	24	23	46	39	43	45	41	48	41	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	11.7	19.6	27.8	31.2	33.3	34.0	32.8	31.5	24.9	18.9	13.0
		MCWB	8.3	7.0	11.1	16.2	18.7	22.7	23.7	22.7	22.1	17.6	14.3	10.4
	2%	DB	6.5	7.8	13.7	23.0	28.0	30.6	31.8	30.5	28.1	21.8	16.2	9.1
		MCWB	3.9	3.8	8.2	13.3	17.5	21.2	22.2	21.5	20.3	16.3	12.6	6.9
	5%	DB	4.0	4.9	10.4	19.7	25.4	28.2	30.2	28.8	25.7	19.0	13.0	6.7
		MCWB	1.7	2.1	5.4	11.7	16.2	19.4	21.4	20.8	19.3	14.8	9.6	4.6
	10%	DB	2.2	2.7	7.7	16.4	22.6	26.2	28.3	27.2	23.4	16.8	10.8	4.3
		MCWB	0.4	0.4	4.1	9.4	14.8	18.4	20.5	19.9	18.0	13.1	7.8	2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	7.9	12.9	17.6	21.0	24.2	25.1	24.2	23.2	19.5	15.7	10.8
		MCDB	10.5	10.8	17.4	23.8	27.7	30.0	31.8	29.7	28.9	22.8	17.7	12.6
	2%	WB	4.4	4.3	8.7	15.1	19.3	22.7	23.7	23.0	21.8	17.3	13.1	7.5
		MCDB	6.2	7.0	12.5	21.5	25.3	28.2	29.5	28.1	26.1	20.3	15.5	9.0
	5%	WB	2.1	2.4	6.1	12.5	17.8	21.5	22.7	22.1	20.5	15.4	10.4	4.9
		MCDB	3.7	4.6	9.9	17.9	23.0	26.0	27.8	26.6	24.1	18.5	12.8	6.6
	10%	WB	0.6	0.8	4.1	10.2	16.2	20.1	21.7	21.2	19.1	13.7	7.9	2.7
		MCDB	1.9	2.3	7.2	15.7	21.1	24.1	26.3	25.2	22.3	16.2	9.9	4.2
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.6	11.3	12.8	13.4	12.7	12.8	12.8	12.6	11.3	9.7	8.9
		MCDBR	10.8	12.6	15.3	19.8	18.3	15.6	14.8	14.3	14.8	14.7	13.7	10.6
		MCWBR	9.1	9.6	10.0	10.8	9.1	7.5	6.7	6.8	7.9	9.2	10.1	8.8
	5% WB	MCDBR	10.1	11.6	13.8	17.0	14.8	13.0	12.8	12.0	12.2	12.7	11.9	10.0
		MCWBR	8.8	9.2	9.6	10.2	8.0	6.9	6.4	6.2	6.9	8.8	9.7	8.8
Clear-Sky Solar Irradiance	taub	0.284	0.288	0.319	0.356	0.384	0.429	0.445	0.420	0.370	0.335	0.308	0.281	
	taud	2.385	2.347	2.390	2.390	2.359	2.246	2.205	2.264	2.391	2.501	2.522	2.474	
	Ebn at Noon	849	910	922	909	888	845	826	835	855	848	821	824	
	Edh at Noon	77	95	104	113	120	135	140	127	104	82	68	65	
All-Sky Solar Radiation	RadAvg	1.58	2.46	3.62	4.46	5.01	5.40	5.60	5.07	4.11	2.53	1.68	1.21	
	RadStd	0.14	0.22	0.23	0.53	0.58	0.49	0.40	0.27	0.28	0.27	0.19	0.12	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.60	N/A	N/A	N/A	N/A	-0.49	N/A	N/A	+130	N/A	(46)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	+32	(47)

Nomenclature: See separate page